

Project Life Cycle, Phases - Project Planning, Project Execution, Project Closure, Project Risks, Project Cost Risk Analysis, Time and Cost overruns

3.1 Introduction

Every programme, project or product has certain phases of development. The different phase of development in an investment proposal or project is called life cycle. A clear understanding of these phases permits entrepreneurs, managers and executives to have better control over existing and potential resources in the achievement of the desire goals.

Project life cycle divides the sequence of operations of project in to different phases. Regardless of scope or complexity, any project goes through a series of stages during its life. Project activities must be grouped into phases to facilitate project manager and his team to plan and organize various inputs effectively. It also helps in identifying deviations and thus helps in decision making with regard to continuation or termination of the project.

3.2 Phases of Project Life Cycle

The project life cycle typically passes through four stages, viz., Initiating, planning, executing and closing. The following figure shows the Project Life Cycle.

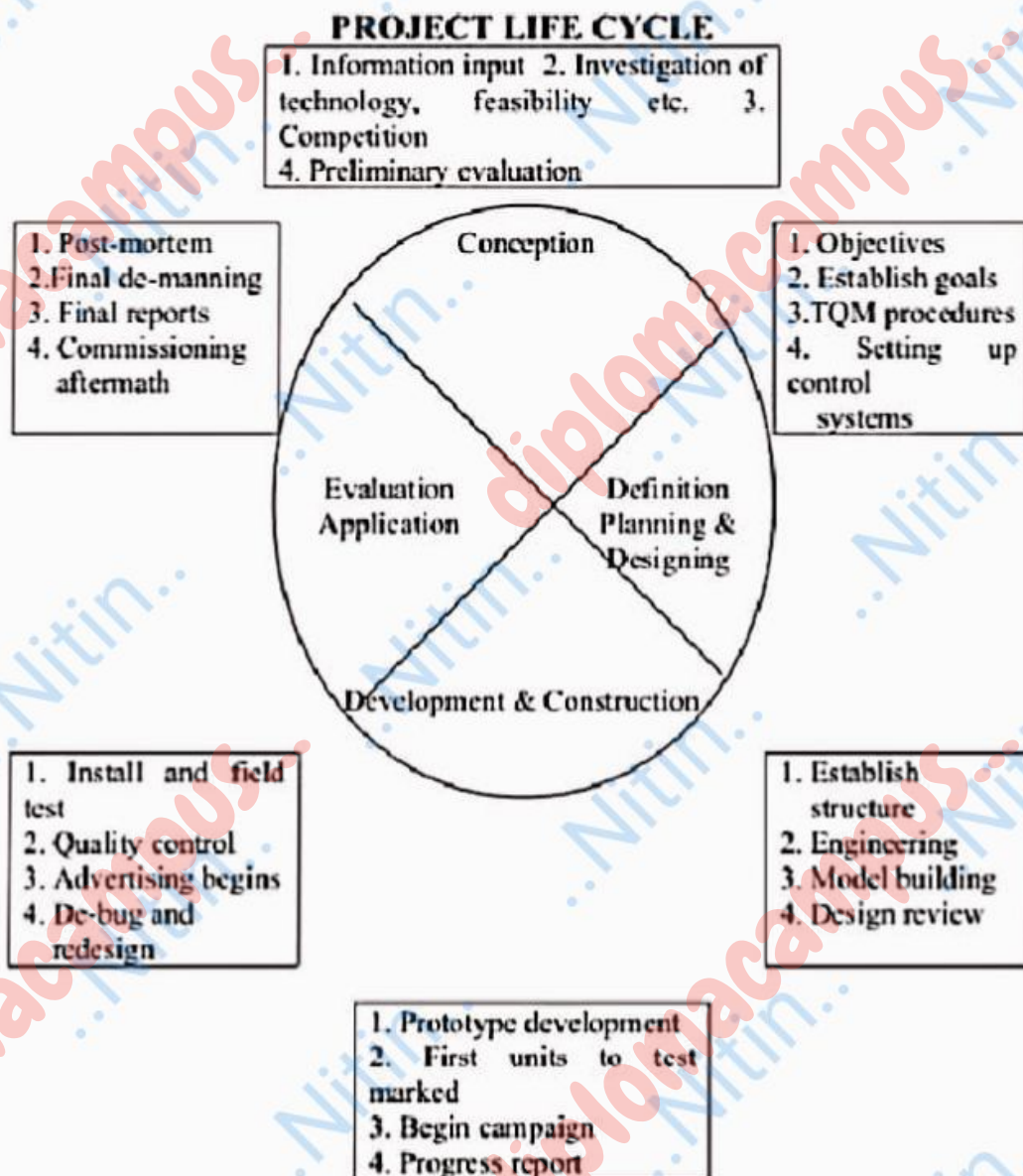
The starting point begins the moment the project is given the go ahead. Project efforts starts slowly, build to a peak and then declines to delivery of the project to the customer. The stages in the project life cycle are:

a) Project Initiation Stage: In this stage, the specifications of the project are defined along with the clear cut project objectives. Project teams are formed and their major responsibilities are assigned. More specifically, this stage defines the goals, specifications, tasks and responsibilities.

b) Project Planning Stage: In this stage, the effort level increases and plans are developed to determine what the project will entail, when it will be scheduled, whom it will benefit, what quality level should be maintained and what the budget will be. More specifically, this stage will include planning schedules, budgets, resources, risks and staffing.

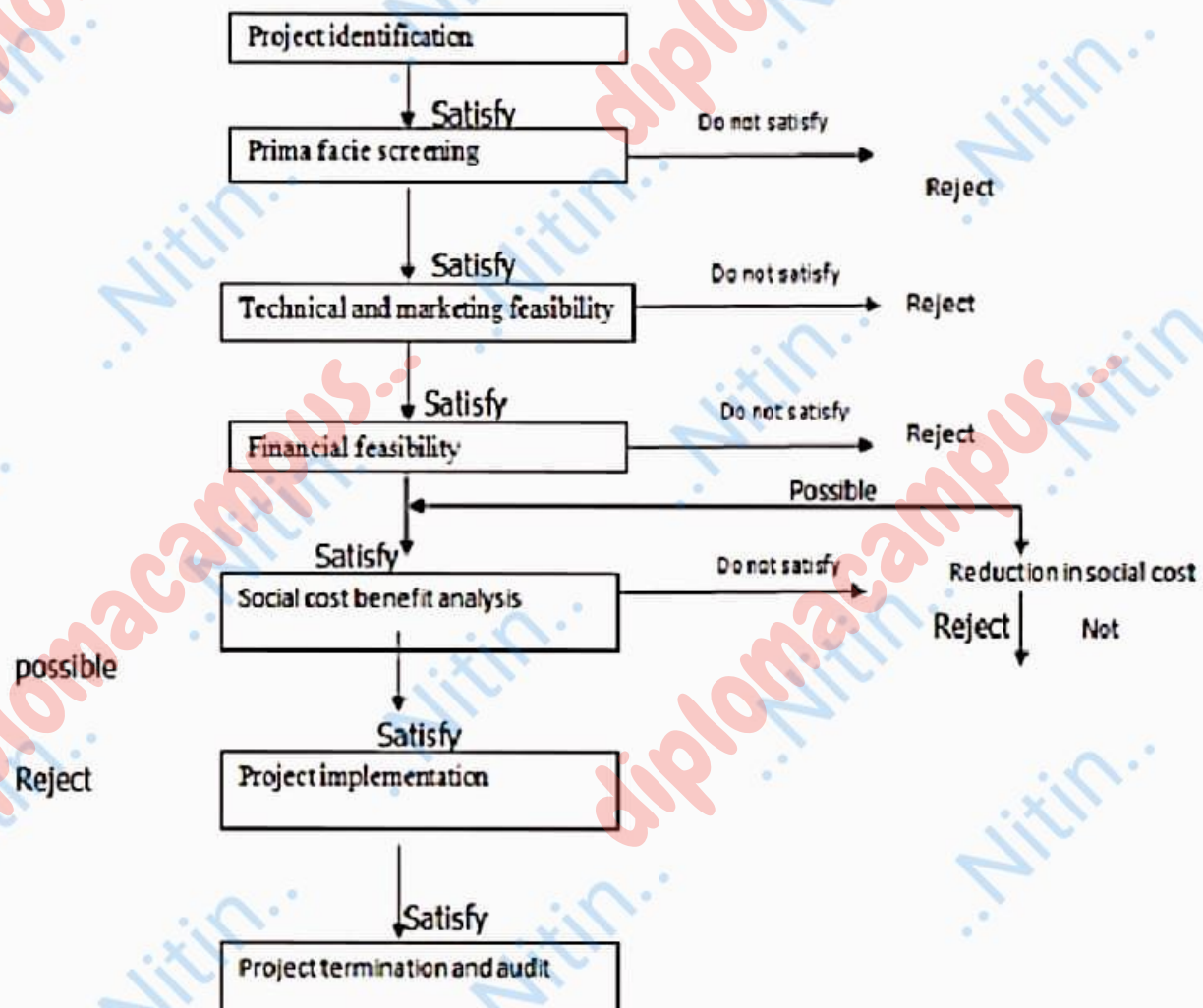
c) Project Execution Stage: In this stage, a major portion of the project work takes place. The physical product is produced (For eg., house, bridge, software program, report, etc). Time, cost and specification measures are used for control. More specifically, this stage will take care of status reports, changes, quality and forecasts.

d) Project Closure stage: This is the final stage which includes two activities, viz., delivering the outcome of the project to the customer and redeploying the project resources. Delivery of the project might include customer training and transferring documents. Redeployment usually involves releasing project equipment/ materials to other projects and finding new assignments for team members. More specially, this stage will undertake activities relating to training the customer, transfer of documents, releasing resources, releasing staff and learning lessons.



3.3 Project Management Life Cycle General

The process of project management starts with project identification. An entrepreneur searching for some investment opportunity is looking everywhere for a project. There are so many consultancy organizations, seminars and journals which are helpful in identification of a project. Figure reveals a general format for the process of project management.



Project identification and prima facie analysis: This step is crucial as 99.2% projects are dropped during this step. To mention in a positive way, a big percentage of the projects which cross the hurdle of prima facie study reaches its destination of implementation. The entrepreneur is afraid during this particular processes, especially when he is a fresh entrepreneur.

- **Performance of existing industries:** It is generally observed that we are looking at the performance of existing industries in the proposed sector. If pesticides industries are not doing well, the idea of setting up a project of pesticides company is dropped.

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- **Price trend:** Increased price levels also reveals increased demand levels and thus it creates scope for a new project.
 - **Price difference between international and domestic prices:** Commodity with lower international price as compared to domestic price has always a danger of imports whereas commodity with higher international prices is lucrative as it provides opportunity to export.
 - **Government policies:** Any sector which is promoted by government is always boosted. Current government support to infrastructure has led to the movement of various industrial groups in the field of infrastructure.
 - **Location aspects:** The entrepreneur also searches the benefit of location with respect to raw material availability, market proximity, government benefits and other infrastructure availability like road, power, and water.
 - **Financial position:** The entrepreneur also analyses the approximate cost of the project during this phase and confirms the availability of funds to proceed to next step.

Project preparation: The projects which have crossed the hurdle of prima facie study are studied in depth and a Detailed Project Report (DPR) is prepared. The various in-depth analyses done in this stage include.

- **Technical feasibility:** This includes the selection of an appropriate technology, source for the selected technology (developed or transferred), capacity planning, location selection, raw material identification etc.
 - **Market feasibility:** During this study, demand supply gap is determined. The prime competitors and their strengths and weaknesses are analyzed, marketing channels are determined and planned, and the target market and competitive strengths are analyzed.
 - **Financial feasibility:** The cost of project and sources of finance are planned. The financial appraisal is done for the project. This is analyzed using various ratios and capital budgeting techniques like Break Even Analysis, Return on Assets, Profit Margins, Net Present Value (NPV), Pay Back Period (PBP), Internal Rate of Return (IRR), etc.
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- **Social cost benefit analysis:** During this particular analysis, we try to determine the cost to the nation due to the proposed project and compare with the benefits.

Project implementation: During this phase, teams are selected, activities are identified and are allotted to group members, and schedules of activities are formed and monitored. This is done with the help of various tools like Gantt chart, Network Diagrams and monitoring tools.

Project review: The review report is a very important document that describes the problems faced and the changes made in the original project plan.

3.4 Project Planning

Project planning phase follows the project initiation phase. Countless hours during the succeeding phases can be saved with proper planning.

The purpose of the project planning phase is to:

- Determine project requirements
- Decide project cost and schedules
- Search for sources of all resources

The basic processes of the project planning phase are:

- ✓ **Defining the scope:** Define the scope of the project and its limitations.
Preparing the work breakdown structure: Divide the whole project into smaller activities
- ✓ **Role assignment:** Assign jobs to individuals or group of individuals as predefined activities or tasks.
- ✓ **Project scheduling:** Determine optimum schedule of the project and show it on a Gantt Chart.
- ✓ **Fund allocation:** Allocation of funds for individual activities

Other subsidiary processes in the planning stage are:

- ✓ **Risk management planning:** It includes identification of possible causes and effect of the risks and trying to reduce the impact of risk.
- ✓ **Procurement planning:** Decisions regarding all products, services or resources needed to accomplish the project.

In the planning stage, various steps are taken which includes:

- ✓ **Final techno-economic feasibility of the project:** This is the last chance for changing the decision, as after this stage, it proves too costly to shut down the project or change the project.
- ✓ **Basic engineering and process design:** The process is selected and basic engineering is done. The documents with respect to equipment specification are prepared.
- ✓ **Division of work/responsibilities:** Different activities are allocated to individuals or groups.
- ✓ **Identify potential vendors and subcontractors:** No project is complete without the help of outside expert agencies called subcontractors. The potential suppliers of various equipment, civil construction agencies and similar agencies are identified and negotiated.
- ✓ **Detailed engineering design:** based on the designs of equipment supplier, detailed engineering is performed. The final layout is prepared and the work schedule prepared.
- ✓ **Final estimation of the cost of the project:** The above steps lead to finalizing quite accurate cost of the project. This is essential as the next step would involve arrangement of funds.
- ✓ **Decision of capital structure and means of finance:** The final decision with respect to financing the project is needed during the planning phase. It is a crucial decision generally taken by core strategic group with the advice of finance managers.
- ✓ **Final schedule of implementation (next phase):** The next phase will be implementation. A proper schedule of implementation is essential to avoid confusions. The schedule of implementation tells all the members of the team when a particular activity should start and end. It will provide the milestones of every activity. The techniques used are PERT, CPM, Gantt chart, crashing resource allocation and resource leveling.

3.5 Project Execution

Project execution is characterized by the actual work on the tasks planned and project control involves the comparison of the actual performance with the planned performance and taking appropriate corrective action to get the desired

output. During this phase, project team is responsible for the following activities:

- The team members perform the tasks allocated in the earlier phase under the supervision of the project manager and report to him.
- Project manager is responsible for performance measurement, which includes finding variances with respect to cost, schedule and scope.
- Project manager is responsible for providing project status report to all key stakeholders. He should specifically inform the deviation from the plan to the stakeholders. He should also determine the root cause for the deviations and suggest the alternate actions to encounter the deviation caused or expected. This helps stakeholders to decide the corrective action to be taken.
- All project key stakeholders are responsible for the review of the variances.
- All project key stakeholders are responsible for taking necessary action of the variances thus determined so as to complete the project within time and cost.

The basic process of the project execution can be:

- ✓ Execution of the project plan
- ✓ Handle the changes
- ✓ Project control

The subsidiary processes during project execution can be:

- ✓ Quality control
- ✓ Performance monitoring
- ✓ Project administration
- ✓ Risk monitoring and control
- ✓ Scope and control
- ✓ Schedule and cost control
- ✓ Management of outside agencies (subcontractors)

The key activities during this phase of execution include:

Award contracts to contractors, vendors, subcontractors: Final selection of suppliers of various supplies of services (generally termed contractors) and physical equipment (generally termed vendors).

Procure equipment and services: After continuously monitoring the suppliers, the project team has to procure the goods and services.

Erection of equipment: The procured equipment needs to be placed on the designed place after preparing the required foundation.

Control and monitor project cost, schedule and scope: As majority of efforts, time and cost are incurred during this phase, it is critical to monitor the project schedule and cost during this phase. This is generally done using various tools like Gantt chart and Earned Value Analysis.

Motivation of project team: As this phase consumes maximum energy of the team members, motivating them during this phase is critical to the success of the project.

3.6 Project Closure

The last step performed to say good bye to a project is the termination phase. The termination of a project is inevitable, but how it is terminated and when may have a profound and long lasting impact on the organization and its employees.

In the end, all projects, both successful and unsuccessful, will have to be terminated. During the termination phase, the project's resources are redistributed, financial records are closed, and project personnel are reassigned. The organization's sensitivity to the concerns of the project team can have a lasting impact on their commitment and productivity. Lastly, a final report, which discusses the project's successes and shortcomings, is prepared for senior management. This report can significantly influence how the organization manages projects in the future.

According to Meredith and Mantel (1995), there are three ways to terminate a project: extinction, inclusion, or integration. Termination by extinction means the project is completed. For example, the new project has been developed and given to the client, the building has been completed and accepted by the purchaser, or the software has been installed and is running.

By contrast, termination by inclusion is a very different process. The complete project team and its equipment are transferred to a new division. As one might expect, this type of change places significant additional stress on the day-to-day

operations of the organization. Project managers and team members must be sensitive to these stresses until the organization is able to settle into a new and more stable routine.

The most common, but also the most complex, method of termination is by integration. The project's resources, personnel, and functions are absorbed as a part of the original organization. The major problem associated with this termination process is the ability of the organization to blend technological differences between the project and the organization. Past experience appears to play a key role in successfully integrating terminated projects.

3.7 Project Risks

Risk management is the process by which risks to the project are formally identified, quantified and managed. A project risk may be identified at any stage of the project by completing a risk form and recording the relevant risk details within the risk register.

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3.8 Types of Risks: Illustrations

There are various sources of risk. Any change in economy of a nation or globe, any change in price levels of inputs or outputs may lead to risk. Let us enumerate the various sources of risk.

Operational risk: Any decrease in operational efficiency may lead to reduced profitability or even at times can lead to loss for the project. In some projects, the effects of decreased operational efficiency may be high or may not be high. Similarly, the possibility of variation of efficiency may vary from project to project. The breakdown of machines, demand and supply of the resources and products, shortfall of the goods and services, lack of efficient logistic and inventory may lead to inefficiency of production. Operational leverage is a measure of operational efficiency.

Market risk: Although every project undergoes market potential analysis and demand forecasting, there are chances of variations due to various foreseen or unforeseen reasons, which may adversely affect market potential or sales of product or services to be rendered by the planned project. This is user acceptance risk.

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Economic risk: Global or national economy is always in different phases and no project can shield itself from ongoing economic changes. All projects are prone to risk of economy. Recession in economy reduces the purchasing power of the customers and thus can affect the sales or revenues of the project. It can also lead to reduce demand as well as selling price. Inflation rate varies with changes in economic situation which also affects prospects of any project.

Financial risk: Any change in interest rates or cost of capital will affect the prospects of a project. A project which is feasible at the expected return of 12% may become infeasible if the expected return increases to 15%. The expected returns directly vary with interest rates. The degree of financial risk for different projects varies with its debt equity ratio and can be measured by the financial leverage.

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Technological risk: The failure of the selected technology is one of the biggest risks to any project. Moreover, a project may face the problem of up gradation of technology. Many projects have failed due to inability to cope up with technological innovations. This particular source of risk is vital in the present world scenario as technology is changing very fast.

Commercial risk: There is always the risk of a customer or debtor becoming insolvent. Although the company does manage and keeps provision for such bad debts, at times when the customer is a major one and the amount outstanding with him is large, it may affect the working of the project.

Quality risk: Today's world is quality conscious and any degraded quality product may lead to rejections and the brand may lose its value in the long run if it is not able to maintain quality norms as per the customer's requirement.

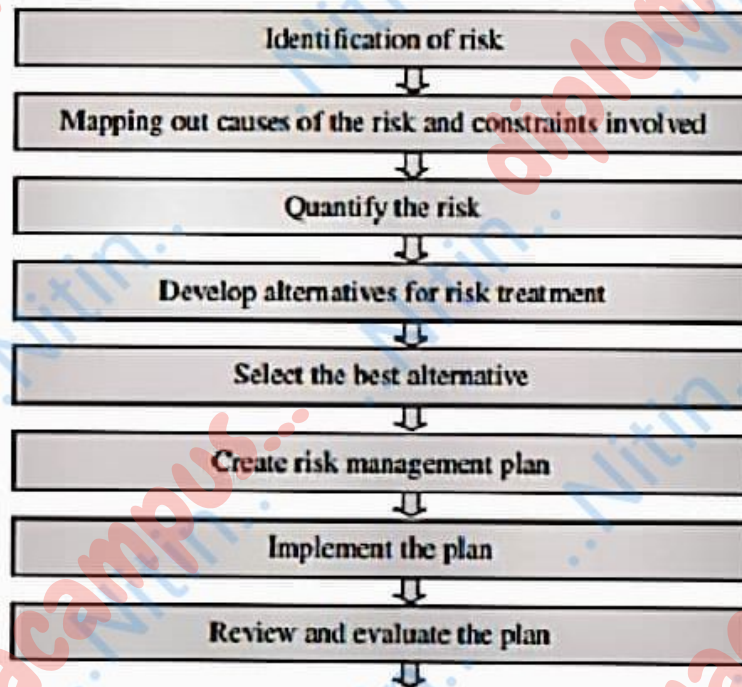
Legal or regulatory risk: Political environment of a country may lead to changes in legal or regulatory changes and may cause risk due to new taxes being imposed or barriers to imports or exports being introduced.

International risk: If international prices with addition of transaction cost (transportation cost and duties) becomes less than domestic prices, there is an import threat. Any change in foreign exchange rates also affects projects with international exposure. A devaluated Indian currency may adversely affect a project which is dependent on imports of goods or services. Similarly, devaluation of foreign exchange may adversely affect project which are export

oriented. Risk of war, imposition of export or import bans and change in political scenario of a country can also lead to risk in project involving international business.

3.9 Risk Assessment Techniques with Illustrations

The various steps in managing risk are as follows:



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Identification of risk: There are various methods used for risk identification. They are:

- ❖ **Objective related risk identification:** The activities of the project are not in coherence with the objectives of the project.
- ❖ **Source based risk identification:** The sources or the stakeholders of the project are not as per expectations.
- ❖ **Industry based risk identifications:** The overall sector of industry may not be in a good shape.
- ❖ **Economy related risk identification:** The project is not fulfilling the desired performance due to poor economic conditions in aggregate.
- ❖ **Time based risk identification:** The project is running far behind the desired schedule.
- ❖ **Financial ratio based risk identification:** This is one of the most common methods of risk identification. Various financial ratios reveal the project getting riskier.

Mapping out causes of risk and constraints involved: Identify the cause/causes of the risk. There can either be one cause or more than one possible causes for the risk involved. The project owner should search as many possible causes as possible. There can be financial, economic, marketing, production or technology constraints. The cause identified may be varied in the boundaries of constraints. For example, there is a high deviation in the cost of production; the cause can be poor efficiency of the organization.

Quantifying the risk: Various methods are used for quantifying the risk.

Develop the alternatives: There can be various methods of reducing the risk, although it is not possible to eliminate it totally. There are various methods of risk management which are as follows:

- ❖ **Mitigation:** It is the process of reducing either the probability or consequence of an event which can cause threat to the project. This includes financial, marketing or other measures.
- ❖ **Sharing:** If a project is expected to have a higher risk, it can be shared with some other firm using joint ventures to reduce the risk.
- ❖ **Outsourcing:** This is a very common methodology used in recent years to reduce the risk. The riskier component is outsourced. Outsourcing can also be very helpful in the situation where huge capital investment is the cause of risk.
- ❖ **Diversification:** Diversification is a very common methodology suggested since ages to reduce the risk. Diversification to a product or project with high negative correlation can reduce the risk of both the projects and products to a great extent.
- ❖ **Abandonment:** If a project is exposed to risk which is not manageable then, the project should be abandoned before its useful life. This is possible when exit is possible.
- ❖ **Retention:** Risk retention is a viable strategy for small risks. All risks that are not reduced or transferred are retained by default. In this situation, the firm should always be in a position to sustain itself in the occurrence of threatening event.

- ❖ **Contingency funding:** Contingencies are reserved for additional cost due to inflation or non-predicted costs. This eliminates sudden funding requirement during project implementation.
- ❖ **Time buffers:** Managers tend to start the activities at the latest start times as a human practice. A too much crashing leads to problem in practical applications. Thus, some time buffers are kept for certain activities.

Selection of best alternative: After developing various alternatives, the next step for risk management is to select the best alternative, which is cheap and easily manageable.

Creation of risk management plan: The selected risk management strategy should always be implemented properly to obtain the result. The project manager should plan the strategy and share with all the stakeholders. Individuals should be assigned responsibilities. In the event of risk retention strategy, guidance should be provided to top management and the operations management team to implement the measures in the situation of occurrence of threatening event. The top management should be well aware of the action plan and the risk involved.

Implementation: The project manager should lead the team to implement the plan. Constraints should always be identified and quantified before the implementation of risk reduction plans. Coordination between the project manager and his team is crucial in the success of the plan.

Review: This is the phase, many a time, avoided by managers. But this is a very important activity as review may help in further improvement (reduction of risk to a greater extent). Moreover, if the desired results are not obtained, then some more steps may be required which are identified during the review phase.

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